

Markowitz Geometry

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September 16, 2026

1 Constraints

Markowitz introduces us to two basic mathematical constraints for asset weights when constructing portfolios.

1. **Budget:** The sum of all the weights w_i inside the portfolio should be equal to 1.

$$\sum_{i=1}^n w_i = 1$$

2. **No Short Selling:** The portfolio does not allow for short selling, investors can only buy securities. Each individual asset weight must be non-negative.

$$w_i \geq 0$$

2 Dimensionality Reduction

Imagine we have a 3 asset portfolio, each of them adds another dimension, making it more complex to analyze. However, we can move from three dimensions to two by leveraging one of the constraints.

We know from the first constraint that the sum of the weights within the portfolio must equal 1. Applying this to our three-asset portfolio gives us:

$$w_1 + w_2 + w_3 = 1$$

We can now solve for w_3 :

$$w_3 = 1 - w_1 - w_2$$

We can see that the value of w_3 depends on the weights assigned to w_1 and w_2 . This allows us to work and visualize the portfolio in a two-dimensional space instead of a three-dimensional one.

3 Isomean curves

Iso-mean curves are parallel straight lines. They represent all portfolios that have exactly the same level of expected return. To demonstrate that they are parallel lines, we can start with the general equation for a portfolio's expected return.

$$E = \sum_{i=1}^n w_i \mu_i$$

Where:

- w_i is the weight of asset i
- μ_i is the expected return of asset i

Applying this equation to our 3-asset portfolio, it becomes:

$$E = w_1 \mu_1 + w_2 \mu_2 + w_3 \mu_3$$

Recall that we can rewrite w_3 in terms of the weights assigned to w_1 and w_2 by substituting it into the expected return equation.

$$E = w_1 \mu_1 + w_2 \mu_2 + (1 - w_1 - w_2) \mu_3$$

Factoring yields:

$$E = w_1(\mu_1 - \mu_3) + w_2(\mu_2 - \mu_3) + \mu_3$$

Since we are working in a two-dimensional plane, we can assign the x-axis to w_1 and the y-axis to w_2 , we can then solve for w_2 in our equation to obtain the slope-intercept form .

$$w_2(\mu_2 - \mu_3) = E - \mu_3 - w_1(\mu_1 - \mu_3)$$

$$w_2 = \frac{E - \mu_3 - w_1(\mu_1 - \mu_3)}{\mu_2 - \mu_3}$$

$$w_2 = \frac{E - \mu_3}{\mu_2 - \mu_3} - w_1 \frac{\mu_1 - \mu_3}{\mu_2 - \mu_3}$$

Upon examining the equation, we see that it follows the slope-intercept form ($y = mx + b$), where:

- $y = w_2$
- $m = -\frac{\mu_1 - \mu_3}{\mu_2 - \mu_3}$
- $x = w_1$
- $b = \frac{E - \mu_3}{\mu_2 - \mu_3}$

Given that the only parameter subject to change is the expected return (E), depending on the investor's preferences, the only variable that will change value is the y-intercept (b), while the slope (m) remains constant. This causes the resulting isomean lines to shift vertically while maintaining their slope, thereby creating lines that are parallel to one another.

4 Isovariance Curves

The isovariance curves are concentric ellipses that map out all portfolios for a specific level of variance. To prove that they are concentric ellipses, we can start with the matrix definition of portfolio variance:

$$\sigma_p^2 = \mathbf{w}^T \mathbf{\Sigma} \mathbf{w}$$

where:

- $\mathbf{w}$ is the vector of portfolio weights
- $\mathbf{\Sigma}$ is the covariance matrix

Recalling that $w_3 = 1 - w_1 - w_2$, we can substitute this value into our weights matrix.

$$\mathbf{w} = \begin{bmatrix} w_1 \\ w_2 \\ 1 - w_1 - w_2 \end{bmatrix}$$

We do the matrix multiplication.

$$\sigma_p^2 = w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + (1 - w_1 - w_2)^2 \sigma_3^2 + 2w_1 w_2 \sigma_{12} + 2w_1 (1 - w_1 - w_2) \sigma_{13} + 2w_2 (1 - w_1 - w_2) \sigma_{23}$$

We can see that there are three components of the equation corresponding to w_3 , which we can expand:

$$(1 - w_1 - w_2)^2 \sigma_3^2 = \sigma_3^2 + w_1^2 \sigma_3^2 + w_2^2 \sigma_3^2 - 2w_1 \sigma_3^2 - 2w_2 \sigma_3^2 + 2w_1 w_2 \sigma_3^2 \quad (1)$$

$$2w_1 (1 - w_1 - w_2) \sigma_{13} = 2w_1 \sigma_{13} - 2w_1^2 \sigma_{13} - 2w_1 w_2 \sigma_{13} \quad (2)$$

$$2w_2 (1 - w_1 - w_2) \sigma_{23} = 2w_2 \sigma_{23} - 2w_1 w_2 \sigma_{23} - 2w_2^2 \sigma_{23} \quad (3)$$

With this, our fully expanded equation becomes:

$$\begin{aligned} \sigma_p^2 &= w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + \sigma_3^2 + w_1^2 \sigma_3^2 + w_2^2 \sigma_3^2 \\ &\quad - 2w_1 \sigma_3^2 - 2w_2 \sigma_3^2 + 2w_1 w_2 \sigma_3^2 + 2w_1 w_2 \sigma_{12} \\ &\quad + 2w_1 \sigma_{13} - 2w_1^2 \sigma_{13} - 2w_1 w_2 \sigma_{13} \\ &\quad + 2w_2 \sigma_{23} - 2w_1 w_2 \sigma_{23} - 2w_2^2 \sigma_{23} \end{aligned} \quad (1)$$

Grouping terms:

$$\begin{aligned} \sigma_p^2 &= (\sigma_1^2 + \sigma_3^2 - 2\sigma_{13})w_1^2 \\ &\quad + (\sigma_2^2 + \sigma_3^2 - 2\sigma_{23})w_2^2 \\ &\quad + 2(\sigma_3^2 + \sigma_{12} - \sigma_{13} - \sigma_{23})w_1 w_2 \\ &\quad + 2(\sigma_{13} - \sigma_3^2)w_1 \\ &\quad + 2(\sigma_{23} - \sigma_3^2)w_2 \\ &\quad + \sigma_3^2 \end{aligned} \quad (2)$$

By observing the equation, we can see that it takes a similar structure to the general form of a quadratic equation:

$$Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0$$

The form of this equation allows us to graph figures such as circles, parabolas, ellipses, and hyperbolas on the Cartesian plane.

With this in mind, we can relate the terms of the portfolio variance equation to those of the quadratic equation:

- $A = (\sigma_1^2 + \sigma_3^2 - 2\sigma_{13})$ with $x = w_1$
- $B = 2(\sigma_3^2 + \sigma_{12} - \sigma_{13} - \sigma_{23})$
- $C = (\sigma_2^2 + \sigma_3^2 - 2\sigma_{23})$ with $y = w_2$
- $D = 2(\sigma_{13} - \sigma_3^2)$
- $E = 2(\sigma_{23} - \sigma_3^2)$
- $F = \sigma_3^2$

Since each isovariance curve represents all portfolios with a specific level of variance, we will set the portfolio variance (σ_p^2) as the fixed parameter in the equation and then set the equation to zero, ultimately obtaining an equation in the general form of a quadratic equation:

$$Ax^2 + Bxy + Cy^2 + Dx + Ey + (F - \sigma_p^2) = 0$$

Recalling that for any equation of this form there is a discriminant $\Delta = B^2 - 4AC$, if the discriminant is less than 0, an ellipse is formed, we can see if this applies to the market.

Assuming a realistic market scenario where no assets are perfectly correlated, the covariance matrix becomes positive definite, which guarantees that:

$$AC - \frac{B^2}{4} > 0$$

Rearranging yields

$$B^2 - 4AC < 0$$

Therefore, forming an ellipse.